DESIGN CODE OF 6-BIT ALU (VERILOG)

/home/dhruv/cao_lab/vivado_proj/cao_lab_u24ev057/rtl/alu_board/alu6_verilog.v [Verilog]

```
1  `timescale 1ns / 1ps
2  //////////////////////////////////////////////////////////////////////////////////
   //
3  // Company:
4  // Engineer:
5  //
6  // Create Date: 10/01/2026 04:51:00 PM
7  // Design Name:
8  // Module Name: alu6_verilog
9  // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 //////////////////////////////////////////////////////////////////////////////////
   //
21
22
23 module alu6_verilog #( parameter w = 6)
24 (
25     input [3:0] op,
26     input [w-1:0] a,
27     input [w-1:0] b,
28     output reg [2*w - 1:0] c,
29     output reg v
30     );
31
32     localparam lim = 1 << (2*w-1)/3;
33
34     wire [2*w -1 : 0] ae , be;
35
36     assign ae = $signed(a);
37     assign be = $signed(b);
38
39     always @(*) begin
40         v <= 1'b0;
41     case(op)
42
43         4'h0 : c <=  ae + be;
44         4'h1 : c <=  ae - be;
45         4'h2 : c <= a*b;
46         4'h3 : begin
```

```
47                if ( b == 0) begin
48                    c <= {{w{1'b1}} , {w{1'b1}}};
49                    v <= 1'b1;
50                    end
51                else begin
52                    c <= a/b;
53                    v <= 1'b0;
54                end
55          end
56          4'h4 : begin
57                if ( b == 0) begin
58                    c <= {{w{1'b1}} , {w{1'b1}}};
59                    v <= 1'b1;
60                    end
61                else begin
62                    c <= ((a%b) + b)%b;
63                    v <= 1'b0;
64                end
65          end
66          4'h5 : begin
67                if ( b == 0) begin
68                    c <= {{w{1'b1}} , {w{1'b1}}};
69                    v <= 1'b1;
70                    end
71                else begin
72                    c <= a%b;
73                    v <= 1'b0;
74                end
75          end
76          4'h6 : c <= ~a;
77          4'h7 : c <= a & b;
78          4'h8 : c <= a ^ b;
79          4'h9 : c <= a >> b;
80          4'ha : c <= a << b;
81          4'hb : c <= (a >> 1) | (a << w-1);
82          4'hc : c <= (a << 1) | (a >> w-1);
83          4'hd : begin
84                if ( ($signed(a) < $signed(-1*lim)) || ( $signed(a) >= lim)) begin
85                    c <= {{w{1'b1}} , {w{1'b1}}};
86                    v <= 1'b1;
87                end
88                else begin
89                    c <= a*a*a;
90                    v <= 1'b0;
91                end
92          end
93          4'he : c <= a*a - b/2;
94          4'hf : c <= b*b - b%17;
95          default : c <= {{w{1'b0}} , {w{1'b0}}};
96      endcase
97
```

```
98
99      end
100
101
102 endmodule
```

DESIGN CODE OF 7-SEGMENT DISPLAY TOP MODULE (VERILOG)

/home/dhruv/cao_lab/vivado_proj/cao_lab_u24ev057/rtl/alu_board/alu_display.v [Verilog]

```
1  `timescale 1ns / 1ps
2  //////////////////////////////////////////////////////////////////////////////////
   //
3  // Company:
4  // Engineer:
5  //
6  // Create Date: 10/14/2026 01:04:27 PM
7  // Design Name:
8  // Module Name: alu_display
9  // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 //////////////////////////////////////////////////////////////////////////////////
   //
21
22
23 module alu_display(
24     input clk,
25     input [3:0] op,
26     input [5:0] a,
27     input [5:0] b,
28     output [6:0] seg,
29     output v,
30     output sign,
31     output dp,
32     output [3:0] an
33 );
34
35     wire [11:0] alu_out;
36     wire alu_overflow;
37     wire [11:0] mag;
38     reg [15:0] bcd;
39     integer i;
40
```

```
    reg [17:0] refresh_counter = 18'd0;
    wire [1:0] digit_select;
    reg [3:0] current_digit;
    reg [3:0] an_reg;
    reg [6:0] seg_reg;

    alu6_verilog #(.w(6)) alu6_inst (
        .op(op),
        .a(a),
        .b(b),
        .c(alu_out),
        .v(alu_overflow)
    );

    assign v = alu_overflow;
    assign sign = alu_out[11];
    assign mag = alu_out[11] ? (-alu_out) : alu_out;

    always @(*) begin
        bcd = 16'd0;
        for (i = 11; i >= 0; i = i - 1) begin
            if (bcd[3:0] >= 5)
                bcd[3:0] = bcd[3:0] + 3;
            if (bcd[7:4] >= 5)
                bcd[7:4] = bcd[7:4] + 3;
            if (bcd[11:8] >= 5)
                bcd[11:8] = bcd[11:8] + 3;
            if (bcd[15:12] >= 5)
                bcd[15:12] = bcd[15:12] + 3;
            bcd = {bcd[14:0], mag[i]};
        end
    end

    always @(posedge clk) begin
        refresh_counter <= refresh_counter + 1'b1;
    end

    assign digit_select = refresh_counter[17:16];

    always @(*) begin
        case (digit_select)
            2'b00: begin
                an_reg = 4'b1110;
                current_digit = bcd[3:0];
            end
            2'b01: begin
                an_reg = 4'b1101;
                current_digit = bcd[7:4];
            end
            2'b10: begin
                an_reg = 4'b1011;
```

```
                current_digit = bcd[11:8];
            end
            2'b11: begin
                an_reg = 4'b0111;
                current_digit = bcd[15:12];
            end
            default: begin
                an_reg = 4'b1111;
                current_digit = 4'd0;
            end
        endcase
    end

    always @(*) begin
        case (current_digit)
            4'h0: seg_reg = 7'b1000000;
            4'h1: seg_reg = 7'b1111001;
            4'h2: seg_reg = 7'b0100100;
            4'h3: seg_reg = 7'b0110000;
            4'h4: seg_reg = 7'b0011001;
            4'h5: seg_reg = 7'b0010010;
            4'h6: seg_reg = 7'b0000010;
            4'h7: seg_reg = 7'b1111000;
            4'h8: seg_reg = 7'b0000000;
            4'h9: seg_reg = 7'b0010000;
            default: seg_reg = 7'b1111111;
        endcase
    end

    assign an = an_reg;
    assign seg = seg_reg;
    assign dp = 1'b1;

endmodule
```

TESTBENCH CODE OF 6-BIT ALU (VERILOG)

/home/dhruv/cao_lab/vivado_proj/cao_lab_u24ev057/tb/alu_board/alu6_tb.v [Verilog]

```
`timescale 1ns / 1ps
//////////////////////////////////////////////////////////////////////////////////
// Company:
// Engineer:
//
// Create Date: 10/01/2026 05:18:34 PM
// Design Name:
// Module Name: alu6_tb
// Project Name:
// Target Devices:
// Tool Versions:
// Description:
```

```
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 //////////////////////////////////////////////////////////////////////////////////
   //
21
22
23 module alu6_tb(
24     );
25
26     parameter w = 6;
27
28     reg  [3:0] op;
29     reg  [w-1:0] a;
30     reg  [w-1:0] b;
31     wire [2*w - 1:0] c;
32     wire v;
33
34     alu6_verilog uut (
35         .op(op),
36         .a(a),
37         .b(b),
38         .c(c),
39         .v(v)
40     );
41
42     initial begin
43         op = 4'h0;
44         a  = 6'h00;
45         b  = 6'h00;
46         #10;
47
48         op = 4'h0; a = 6'd25;  b = 6'd15;  #10;
49         op = 4'h0; a = 6'hCE;  b = 6'd20;  #10;
50         op = 4'h0; a = 6'h7F;  b = 6'd1;   #10;
51         op = 4'h0; a = 6'hFF;  b = 6'd1;   #10;
52
53         op = 4'h1; a = 6'd50;  b = 6'd20;  #10;
54         op = 4'h1; a = 6'd20;  b = 6'd50;  #10;
55         op = 4'h1; a = 6'h80;  b = 6'h7F;  #10;
56
57         op = 4'h2; a = 6'd12;  b = 6'd10;  #10;
58         op = 4'h2; a = 6'hF4;  b = 6'd10;  #10;
59         op = 4'h2; a = 6'h80;  b = 6'h80;  #10;
60         op = 4'h2; a = 6'h7F;  b = 6'h7F;  #10;
61
62         op = 4'h3; a = 6'd25;  b = 6'd5;   #10;
63         op = 4'h3; a = 6'hE7;  b = 6'd4;   #10;
```

```
64          op = 4'h3; a = 6'd25;  b = 6'd0;   #10;
65          op = 4'h3; a = 6'd0;   b = 6'd7;   #10;
66
67          op = 4'h4; a = 6'd17;  b = 6'd5;   #10;
68          op = 4'h4; a = 6'hEF;  b = 6'd5;   #10;
69          op = 4'h4; a = 6'd17;  b = 6'd0;   #10;
70
71          op = 4'h5; a = 6'd17;  b = 6'd5;   #10;
72          op = 4'h5; a = 6'hEF;  b = 6'd5;   #10;
73          op = 4'h5; a = 6'd17;  b = 6'd0;   #10;
74
75          op = 4'h6; a = 6'h00; b = 6'h00; #10;
76          op = 4'h6; a = 6'hFF; b = 6'h00; #10;
77          op = 4'h6; a = 6'h80; b = 6'h00; #10;
78
79          op = 4'h7; a = 8'h55; b = 8'h0F; #10;
80          op = 4'h7; a = 8'hFF; b = 8'h55; #10;
81          op = 4'h7; a = 8'h00; b = 8'hFF; #10;
82
83          op = 4'h8; a = 8'h55; b = 8'h0F; #10;
84          op = 4'h8; a = 8'hFF; b = 8'h55; #10;
85          op = 4'h8; a = 8'hAA; b = 8'hAA; #10;
86
87          op = 4'h9; a = 6'h61; b = 6'd0;  #10;
88          op = 4'h9; a = 6'h61; b = 6'd1;  #10;
89          op = 4'h9; a = 6'h61; b = 6'd7;  #10;
90          op = 4'h9; a = 6'hFF; b = 6'd6;  #10;
91
92          op = 4'hA; a = 6'h61; b = 6'd0;  #10;
93          op = 4'hA; a = 6'h01; b = 6'd7;  #10;
94          op = 4'hA; a = 6'h01; b = 6'd6;  #10;
95
96          op = 4'hB; a = 6'h61; b = 6'd0;  #10;
97          op = 4'hB; a = 6'h01; b = 6'd0;  #10;
98          op = 4'hB; a = 6'hFF; b = 6'd0;  #10;
99
100         op = 4'hC; a = 6'h61; b = 6'd0;  #10;
101         op = 4'hC; a = 6'h40; b = 6'd0;  #10;
102         op = 4'hC; a = 6'hFF; b = 6'd0;  #10;
103
104         op = 4'hD; a = 6'd0;   b = 6'd0; #10;
105         op = 4'hD; a = 6'd3;   b = 6'd0; #10;
106         op = 4'hD; a = 6'hFD;  b = 6'd0; #10;
107         op = 4'hD; a = 6'd31;  b = 6'd0; #10;
108         op = 4'hD; a = 6'd32;  b = 6'd0; #10;
109         op = 4'hD; a = 6'hE0;  b = 6'd0; #10;
110         op = 4'hD; a = 6'hDF;  b = 6'd0; #10;
111
112         op = 4'hE; a = 6'd12;  b = 6'd5;   #10;
113         op = 4'hE; a = 6'hF4;  b = 6'hFB;  #10;
114         op = 4'hE; a = 6'h60;  b = 6'h7F;  #10;
```

```
115
116         op = 4'hF; a = 6'd3;   b = 6'd5;   #10;
117         op = 4'hF; a = 6'hFD;  b = 6'd5;   #10;
118         op = 4'hF; a = 6'd0;   b = 6'd5;   #10;
119
120         #10;
121         $finish;
122     end
123
124 endmodule
```

CONSTRAINT FILE FOR BASYS 3 BOARD (XDC)

/home/dhruv/cao_lab/vivado_proj/cao_lab_u24ev057/constraints/alu/alu6.xdc [XDC]

```
1  #    input [3:0] op,
2  #    input [w-1:0] a,
3  #    input [w-1:0] b,
4  #    output reg [2*w - 1:0] c,
5  #    output reg v
6
7  set_property PACKAGE_PIN W5 [get_ports clk]
8      set_property IOSTANDARD LVCMOS33 [get_ports clk]
9
10 # Switches
11 set_property PACKAGE_PIN V17 [get_ports {op[0]}]
12     set_property IOSTANDARD LVCMOS33 [get_ports {op[0]}]
13 set_property PACKAGE_PIN V16 [get_ports {op[1]}]
14     set_property IOSTANDARD LVCMOS33 [get_ports {op[1]}]
15 set_property PACKAGE_PIN W16 [get_ports {op[2]}]
16     set_property IOSTANDARD LVCMOS33 [get_ports {op[2]}]
17 set_property PACKAGE_PIN W17 [get_ports {op[3]}]
18     set_property IOSTANDARD LVCMOS33 [get_ports {op[3]}]
19 set_property PACKAGE_PIN W15 [get_ports {a[0]}]
20     set_property IOSTANDARD LVCMOS33 [get_ports {a[0]}]
21 set_property PACKAGE_PIN V15 [get_ports {a[1]}]
22     set_property IOSTANDARD LVCMOS33 [get_ports {a[1]}]
23 set_property PACKAGE_PIN W14 [get_ports {a[2]}]
24     set_property IOSTANDARD LVCMOS33 [get_ports {a[2]}]
25 set_property PACKAGE_PIN W13 [get_ports {a[3]}]
26     set_property IOSTANDARD LVCMOS33 [get_ports {a[3]}]
27 set_property PACKAGE_PIN V2 [get_ports {a[4]}]
28     set_property IOSTANDARD LVCMOS33 [get_ports {a[4]}]
29 set_property PACKAGE_PIN T3 [get_ports {a[5]}]
30     set_property IOSTANDARD LVCMOS33 [get_ports {a[5]}]
31 set_property PACKAGE_PIN T2 [get_ports {b[0]}]
32     set_property IOSTANDARD LVCMOS33 [get_ports {b[0]}]
33 set_property PACKAGE_PIN R3 [get_ports {b[1]}]
34     set_property IOSTANDARD LVCMOS33 [get_ports {b[1]}]
35 set_property PACKAGE_PIN W2 [get_ports {b[2]}]
36     set_property IOSTANDARD LVCMOS33 [get_ports {b[2]}]
37 set_property PACKAGE_PIN U1 [get_ports {b[3]}]
```

```
    set_property IOSTANDARD LVCMOS33 [get_ports {b[3]}]
set_property PACKAGE_PIN T1 [get_ports {b[4]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {b[4]}]
set_property PACKAGE_PIN R2 [get_ports {b[5]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {b[5]}]


# LEDs
set_property PACKAGE_PIN U16 [get_ports {v}]
    set_property IOSTANDARD LVCMOS33 [get_ports {v}]
set_property PACKAGE_PIN E19 [get_ports {sign}]
    set_property IOSTANDARD LVCMOS33 [get_ports {sign}]

set_property PACKAGE_PIN W7 [get_ports {seg[0]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {seg[0]}]
set_property PACKAGE_PIN W6 [get_ports {seg[1]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {seg[1]}]
set_property PACKAGE_PIN U8 [get_ports {seg[2]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {seg[2]}]
set_property PACKAGE_PIN V8 [get_ports {seg[3]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {seg[3]}]
set_property PACKAGE_PIN U5 [get_ports {seg[4]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {seg[4]}]
set_property PACKAGE_PIN V5 [get_ports {seg[5]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {seg[5]}]
set_property PACKAGE_PIN U7 [get_ports {seg[6]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {seg[6]}]
set_property PACKAGE_PIN V7 [get_ports dp]
    set_property IOSTANDARD LVCMOS33 [get_ports dp]
set_property PACKAGE_PIN U2 [get_ports {an[0]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {an[0]}]
set_property PACKAGE_PIN U4 [get_ports {an[1]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {an[1]}]
set_property PACKAGE_PIN V4 [get_ports {an[2]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {an[2]}]
set_property PACKAGE_PIN W4 [get_ports {an[3]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {an[3]}]
```

SCREENSHOT OF ALU AND DISPLAY MODULE CODE

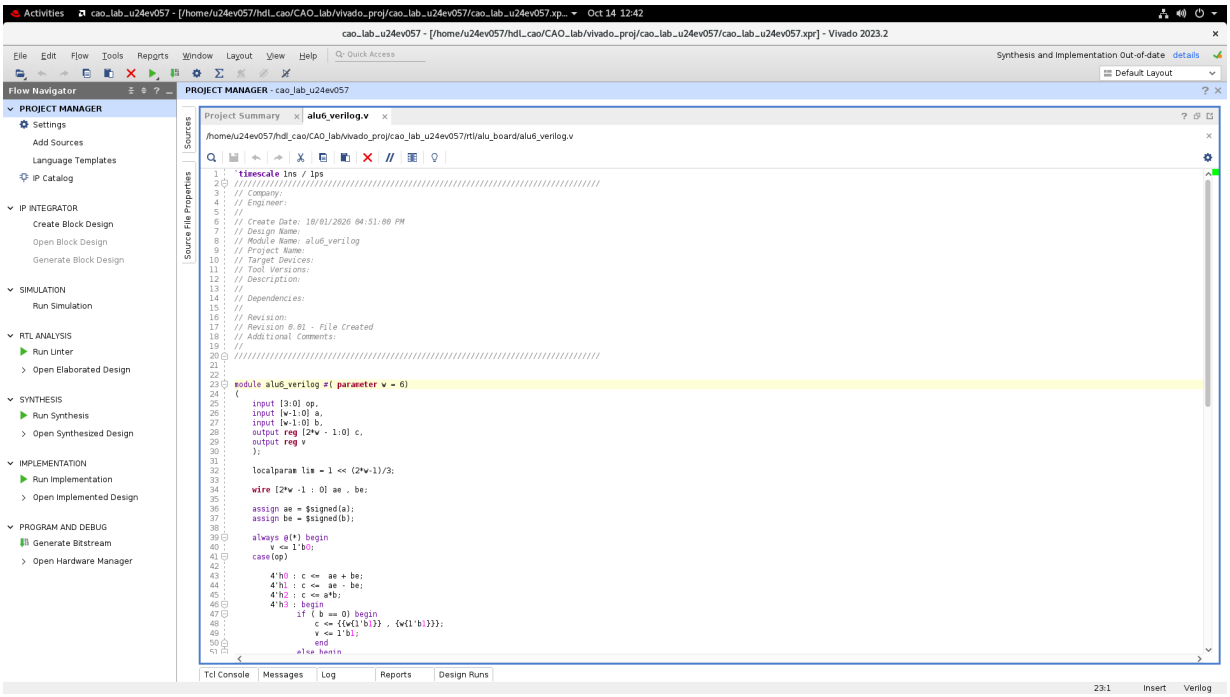


Figure 1: 6-bit ALU Verilog Design Code

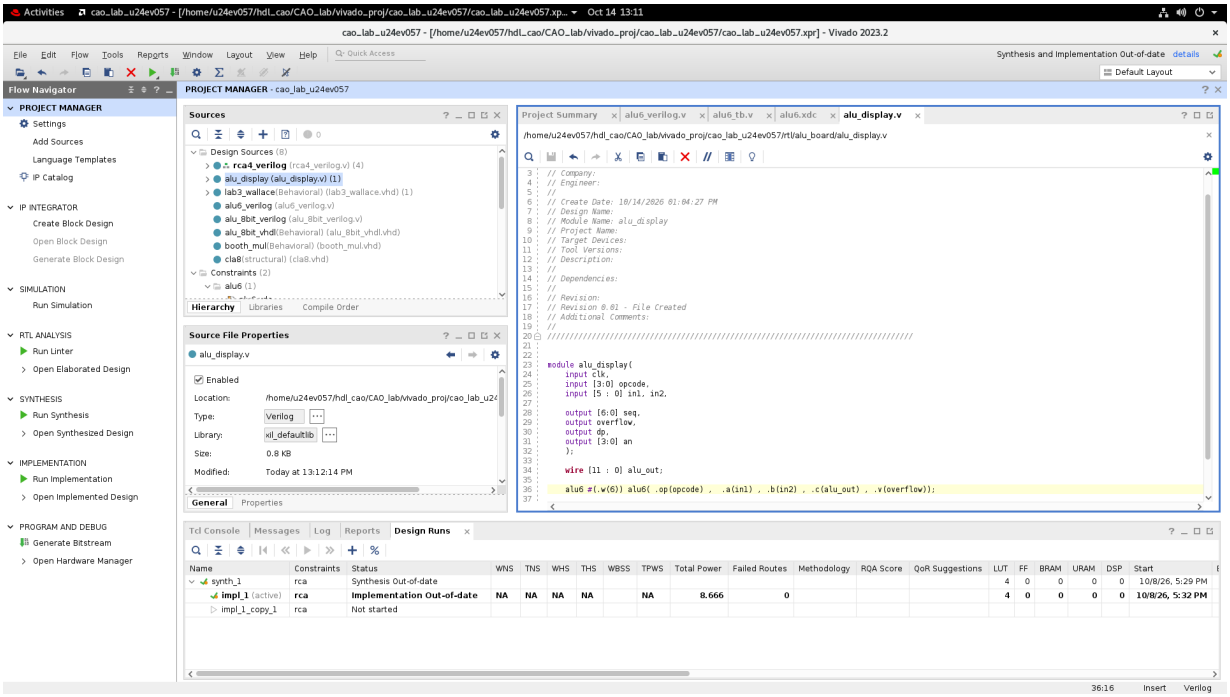


Figure 2: 7-Segment Display Top Module Code

SCREENSHOT OF TESTBENCH AND CONSTRAINTS

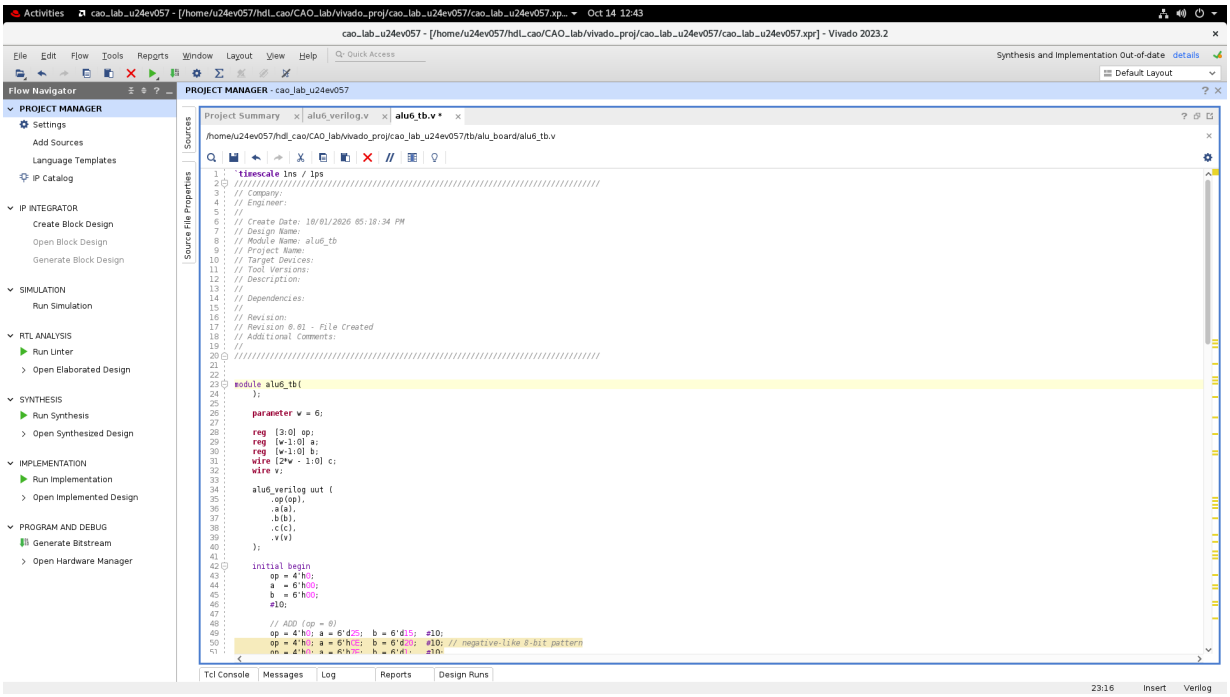


Figure 3: 6-bit ALU Testbench Code

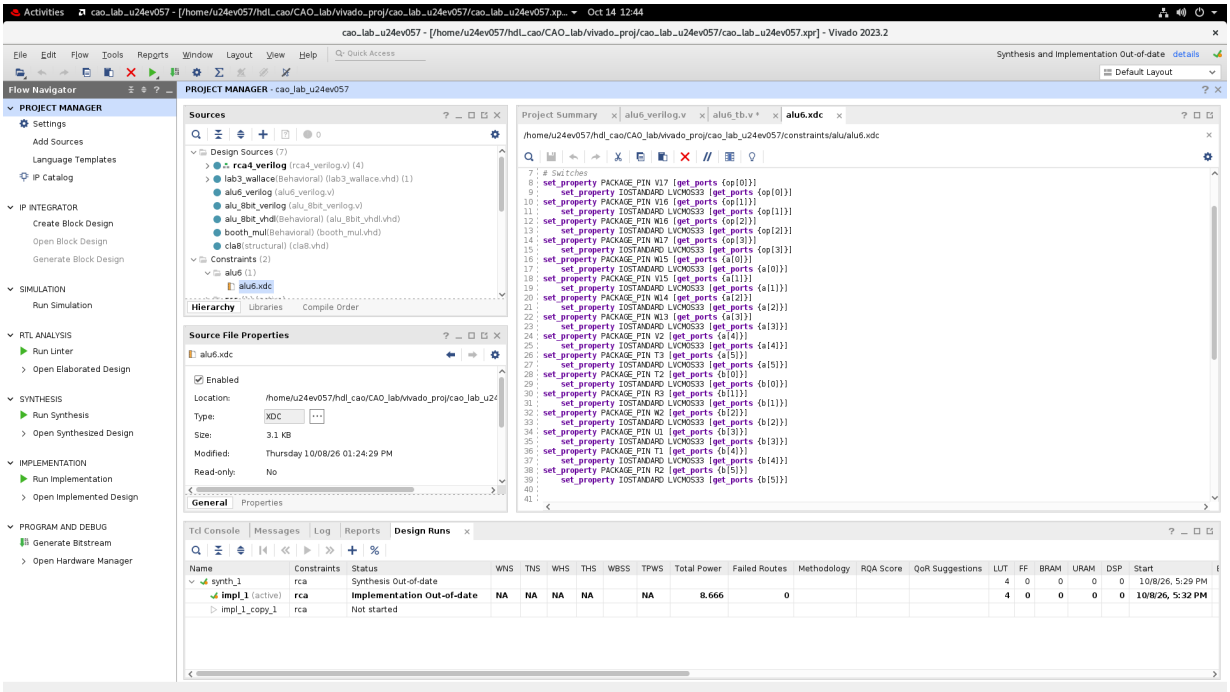


Figure 4: Basys 3 Constraints File

SIMULATION WAVEFORM

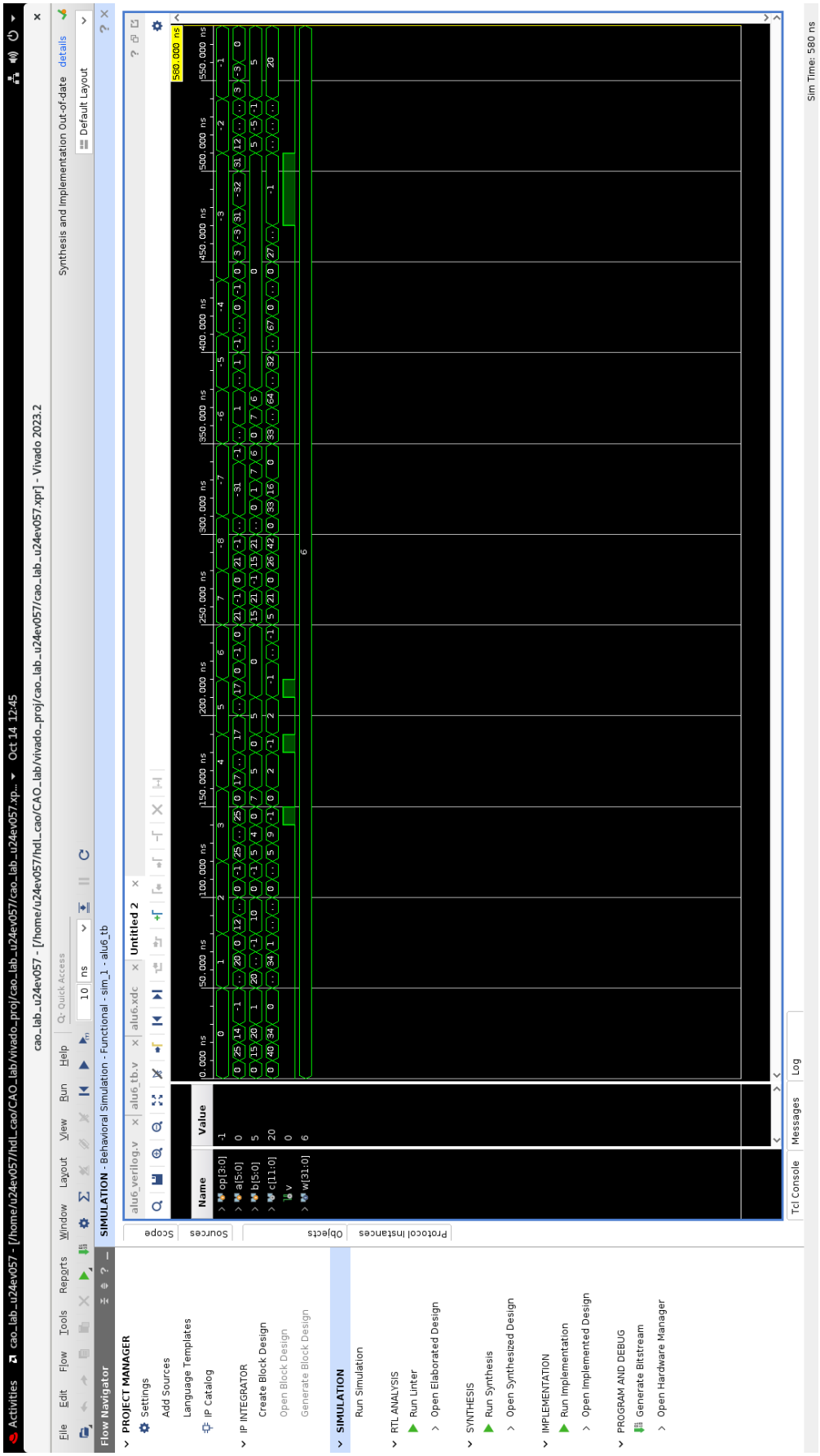


Figure 5: 6-bit ALU Simulation Waveform